

Reimagining Animation Making through Style Transfer

Sujin Kim

School of Art, Arizona State University, Tempe, Arizona, United States
SujinKim@asu.edu

ABSTRACT

"Dissolution" and "Cunabula" are experimental animation projects that explore the potential of artistic style transfer with neural networks in animation filmmaking. The objective is to discover novel production methods using fast, arbitrary, and stroke-adjustable style transfer networks that allow for unconventional technical approaches during the production process and the creation of unique and unexplored aesthetics in moving images powered by deep learning. These projects exemplify a creative approach to leveraging existing neural style transfer techniques to create compelling animations that go beyond mere passive pastiches of certain artistic styles, thereby advancing the aesthetics of contemporary animation.

CCS CONCEPTS

• **Computing methodologies** → Computer graphics; Animation; Machine learning; Machine learning algorithms.

KEYWORDS

Neural Style Transfer, Experimental Animation

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1 INTRODUCTION

Style transfer is a computer vision technique that employs machine learning to synthesize images by finding an image that simultaneously matches the content representation of a photograph and the style representation of a respective piece of art [Gatys et al., 2016]. Since its introduction by Gatys et al. in 2016, machine learning-based techniques for image stylization have garnered significant attention in both research and casual, creativity-targeted image applications. With the rapid advancement of deep convolutional neural networks, style transfer can now be accomplished quickly and with better resolution, enabling real-time stylization using any content-style image pair [Ghiasi et al., 2017]. Subsequent research has led to interactive adjustments in stroke granularity, direction, and intensity, evolving neural style transfers (NSTs) into

art-directable image-stylization tools that go beyond mere one-shot style replication [Reimann et al., 2022].

However, it remains uncertain whether style transfer has been fully recognized and implemented as a serious artistic tool for conveying artistic intentions, enhancing subject matter, and advancing artistic expressions. This paper presents two animated films, "Dissolution" and "Cunabula," as examples illustrating how deep learning algorithms for image stylization can not only enhance a filmmaker's message but also assist artists in their art direction. This approach transcends recent trends in using style transfer applications for casual creativity or pastiche creations.

2 ANIMATION "DISSOLUTION"

2.1 A Multi-Style, Multi-Media Animation Production Method

Traditional animation production typically demands a predetermined set of tools, software, and artistic styles right from the outset to ensure a consistent and cohesive look for the animation. While this approach enhances the sustainability and stability of the production system, it often imposes restrictions on an artist's technical flexibility during production. The production method used in creating the animation "Dissolution" represents a unique departure from the conventional approach to animation. "Dissolution" is the result of a seamless blend of AI-generated animation based on text input (an example in Figure 1), hand-drawn animation (an example in Figure 2), 3D animation (an example in Figure 3), and live videos. These disparate styles of moving images, each characterized by distinct animation tools and techniques, blend harmoniously into a unified, entirely hand-painted animation style with consistent brushwork and tone. This effect is achieved through the implementation of neural style transfer.

The project utilizes a pre-trained model [Google, 2017] for fast arbitrary image stylization, sourced from TensorFlow Hub [TensorFlow Hub, 2023], and incorporates open-source code [Sundaresan, 2021] based on [Ghiasi et al., 2017]. The code was executed using Jupyter Notebook [Jupyter Notebook, 2023], while Gradio [Gradio, 2023] was employed to create an interactive style transfer application and test it in a web browser [Wycliffe, 2022]. The application features user-friendly widgets and customizable UI components for uploading both content and style images, as well as controlling stylization strength. This enables rapid image loading and the download of newly generated stylized images. The integration of the style transfer application in the production of "Dissolution" not only facilitated the inclusion of various animation tools, techniques, and styles at any stage of production but also provided the artist with greater creative freedom.

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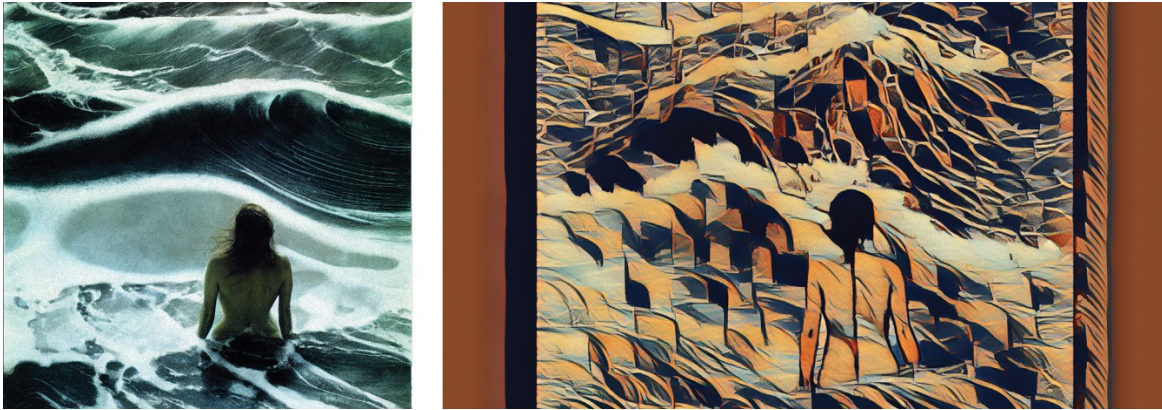


Figure 1: Left: The original AI-generated animation based on text input. | Right: The final animation stylized through the fast-arbitrary style transfer.

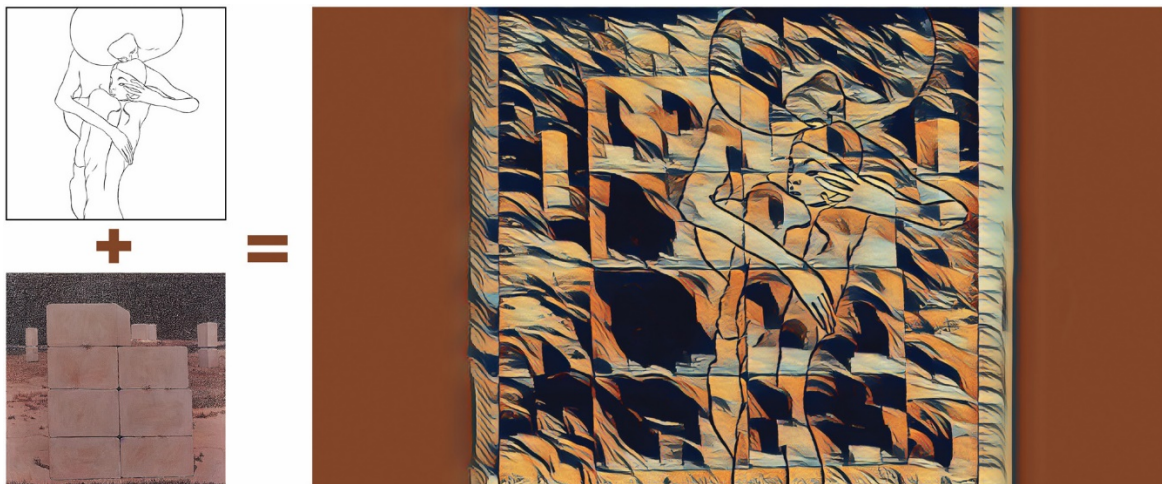


Figure 2: Left: The original hand-drawn animation (Top) and AI-generated animation based on text input (Bottom). | Right: The final animation stylized through the fast-arbitrary style transfer.

2.2 Enhancing Artistic Message through Style Transfer

In the realm of filmmaking, a dissolve is a well-recognized technique that facilitates a gradual transition from one image to another. Typically, it serves as a bridge between various stylistic shots. However, through the application of machine learning-based stylization processes, these traditional dissolves undergo a remarkable metamorphosis, evolving into captivating animated transitions enriched with distinctive morphing effects (an example in Figure 4). These unique animated transitions seamlessly bridge personal and societal imagery within “Dissolution,” serving as an amplification of the artwork’s intended artistic message. They act as a powerful visual metaphor, symbolizing the intricate interplay between an individual’s psychological turmoil and the unpredictability of our turbulent era, where each factor becomes both the cause and consequence of the other—constituting the central theme of “Dissolution.” The fusion of style transfer with established film editing techniques

not only explores uncharted territories in visual representation but also elevates the conveyance of artistic messages to new heights.

3 ANIMATION “CUNABULA”

The Successful implementation of style transfer to enhance the artistic message of the non-narrative animation “Dissolution” has expanded the artist’s artistic practice into the realm of utilizing deep learning algorithms to support art direction in a narrative animated film dedicated to amplifying the voices of historically and socially marginalized people. The animated film, “Cunabula,” demonstrates how style transfer can significantly contribute to establishing the intended artistic mood and atmosphere, fostering a deeper emotional connection with the audience, particularly through deliberate control over brushstroke styles.

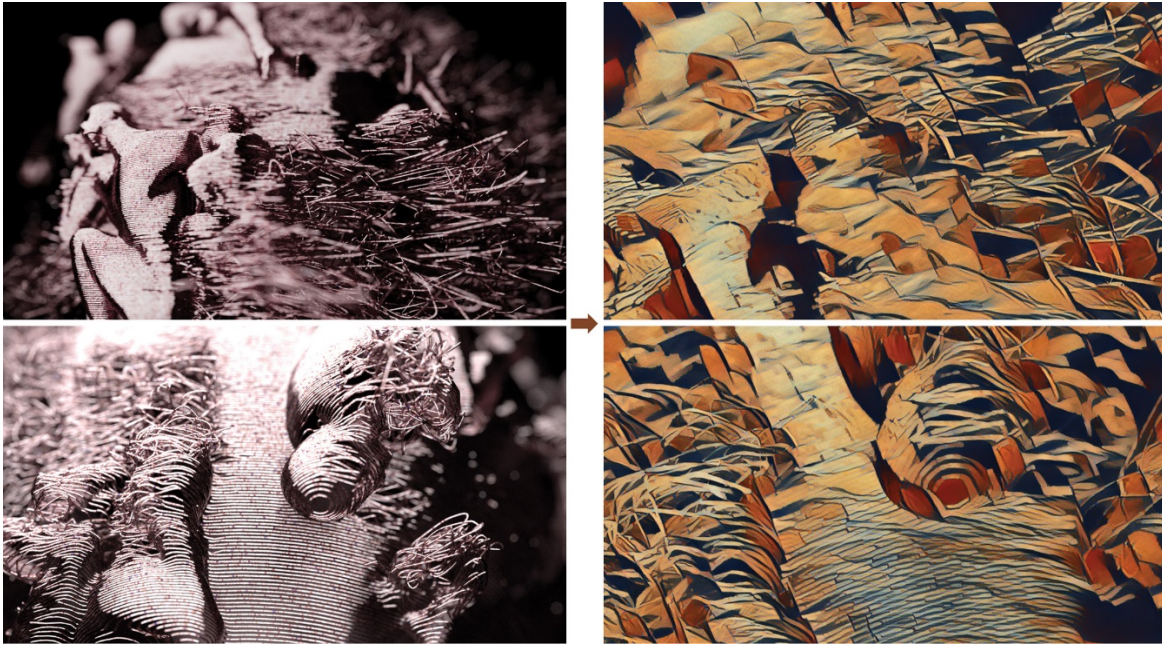


Figure 3: Left: The original 3D animation made using Houdini. | Right: The final animation stylized through the fast-arbitrary style transfer.



Figure 4: From left to right: A transition scene depicting the seamless transformation of an ocean scene into the likeness of a man's face.

3.1 A Real Story

In the aftermath of the Korean War (1950–1953), thousands of mixed-race babies were born to Western soldier fathers and Korean mothers who worked as sex workers in so-called camptowns, which refer to the towns surrounding military bases of United States Forces Korea. Most of these mothers came from impoverished rural families and engaged in the sex industry to provide financial support to their families. Only a small number of mixed-race children were taken back to the United States by their Western fathers, while the majority were placed in orphanages, despite having memories of living with their biological parents. This unfortunate outcome was primarily the result of the harsh realities of the camptowns and the social stigmas associated with being considered 'racially impure' in their native country. As a result, Korean mixed-race babies were systematically 'exported' to Western countries, primarily the United States, through intercountry adoption programs. The adoption of Korean babies by American families was widely embraced, both by those seeking to adopt and by the U.S. government, which

aimed to bolster its liberal image in the new Cold War environment. Similarly, Korean society supported this practice as a means of preserving its homogeneity.

3.2 Advancing Art Direction through Style Transfer

Through fragmented and distorted flashbacks to their childhood with their biological parents and incomplete memories of a mixed-race adoptee, the animated film "Cunabula" portrays the under-represented history of South Korean camptown women and their mixed-race children, as well as the compelling emotional challenges faced by the mixed-race adoptee. The brushstroke styles and feels they deliver in the film play a significant role in crafting the appropriate emotional and sensory experience for the audience. "Cunabula" employs deep learning algorithms for image stylization that incorporates organic details resembling gouache brushstrokes (examples in Figure 5) that can evoke appropriate emotional and



Figure 5: Stills of “Cunabula” that resemble Gouache brushstrokes.



Figure 6: A still of “Cunabula.”

sensory experience for the audience. These images possess a physically tactile quality designed to evoke warmth and nostalgia by embracing visuals reminiscent of childlike paintings, with a strong emphasis on a tangible and hand-painted feel (an example in Figure 6).

The original scenes of the animation were exclusively created using Autodesk Maya, a 3D animation software. Motion capture data was incorporated into animating scenes to achieve lifelike camera and character movements. The final 3D scenes were rendered with Toon shaders, featuring distinct outlines and simple colors to avoid an unintended shading effects after stylization (examples in Figure 7). This approach has enabled neural style transfer network to efficiently extract content representation from the provided content images. For stylization, a stroke-adjustable image-stylization tool built upon [Reimann et al., 2022] and a series of open-source code [Reimann, 2022] has been implemented. Compared to the image

stylization tool used for the project “Dissolution,” where control of the texture distribution in final stylized images could only be achieved by adjusting the scale of input context images or style images, the stroke-adjustable image-stylization tool allows for interactive control over style elements such as stroke granularity, orientation, or intensity. Sophisticated adjustments over brushstrokes during stylizations have been critical elements in advancing the usage of style transfer in “Cunabula” to depict the character’s internal status and complex emotional scenes (as shown in Figure 8). The art-directable image-stylization tool that enables flexible and interactive control over the feel, direction, and intensity of strokes has improved overall quality of art direction in animation filmmaking.

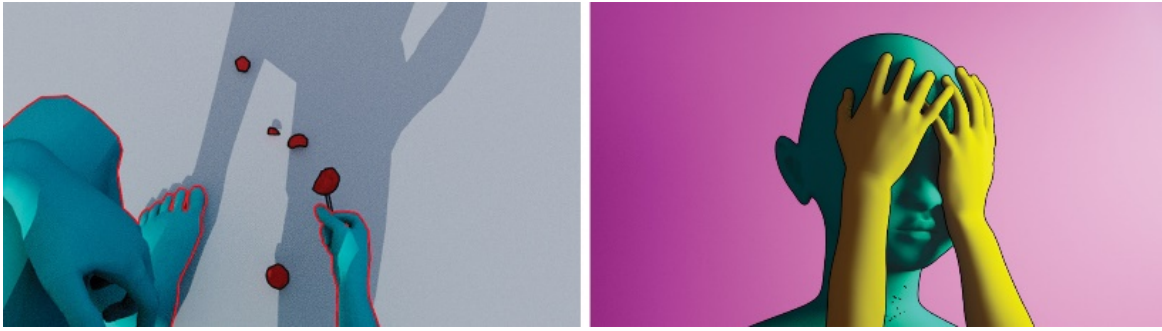


Figure 7: 3D animation stills rendered in Autodesk Maya before style transfer.

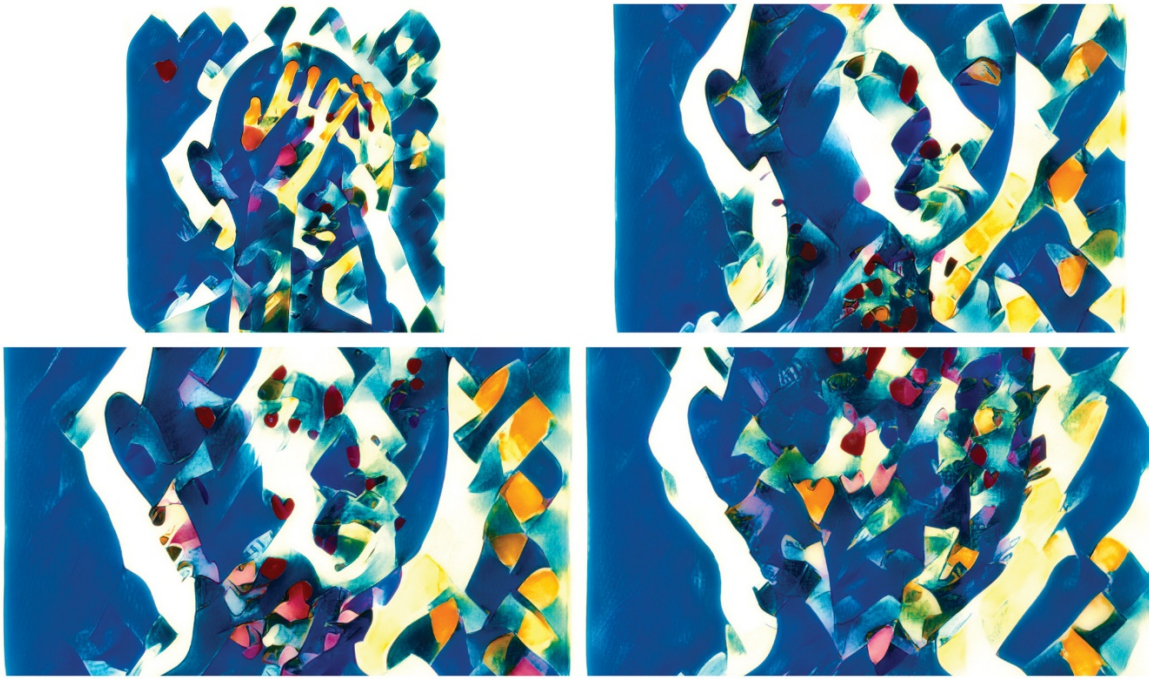


Figure 8: Stills of a scene in “Cunabula” depicting a child’s internal status.

4 CONCLUSION

Both animation projects, “Dissolution” and “Cunabula,” originated from a central question: Can neural style transfer, typically associated with casual art or pastiche creation using user-end applications, be effectively employed to create artistic expressions that align with intended artistic direction and message of an artwork? “Dissolution” serves as a prime example, showcasing that the moving images, empowered by deep learning algorithms for image stylization, can function as unique animated transition effects that play a pivotal role in visualizing the intended message of an animation. Furthermore, “Cunabula” demonstrates how art-directable neural style transfers can be implemented to craft dedicated visual styles that enhance the sense of innocence and nostalgia associated with childhood in an animated film.

However, despite the artistic achievements achieved by implementing neural style transfer techniques in substantial animation projects, several artistic challenges have become apparent. Neural style transfer can sometimes struggle to faithfully reproduce the content of the source animation frames, which poses difficulties in achieving precise control over image details. Moreover, the process of neural style transfer often demands extensive parameter adjustments due to the high unpredictability of outcomes. Furthermore, an overreliance on pre-training data can become problematic, especially when the training dataset is limited or biased. Paradoxically, these artistic constraints have injected the creative process with vibrancy and a commitment to extensive experimentation in crafting desired visual outcomes that align with artistic intent. Ultimately, these challenges underscore the enduring necessity, even in the age of AI art, for deliberate and sophisticated artistic direction by

human creators and their pivotal role as primary investigators in the artistic process.

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